TEXAWAVE ERP • CORE ARCHITECTURE DOCUMENTATION

Finance Module Database Architecture & ER Specification

Visual Entity-Relationship diagrams, foreign key constraint matrix, multi-tenant isolation audit, and cross-schema integration specifications verified against PostgreSQL migrations and Prisma schema.

11 FINANCE MODELS | 192 SCALAR COLUMNS | 32 FOREIGN KEY CONSTRAINTS | 13 UNIQUE CONSTRAINTS | 20 SECONDARY INDEXES

SCHEMA-DERIVED & MIGRATION-VERIFIED

DOCUMENT ID **TEXA-FIN-ERD-2026-10-27**	ACTIVE GIT BRANCH **feature/finance**	PRIMARY SOURCE OF TRUTH **schema.prisma & Migrations**	POSTGRESQL SCHEMA **@@schema("finance")**
PUBLICATION DATE **29 October 2026**	VERIFICATION LEVEL **100% Migration DDL Parity**	COMPLIANCE MANDATES **GAAP Double-Entry • Indian GST**	TENANCY ARCHITECTURE **Multi-Tenant (org_id RESTRICT)**

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1. Executive Overview & Notation Guide

CROW'S FOOT REFERENCE

Every entity, attribute, foreign key constraint, cardinality indicator, and cascade action in this specification reflects the **ACTUAL, CURRENT** database architecture verified directly against `packages/database/prisma/schema.prisma` and the PostgreSQL migrations.

INVIOLABLE ARCHITECTURAL RULES

- **Multi-Tenant Isolation:** All 11 tables strictly reside in `"finance"`. Every table has an `organization_id` foreign key referencing `platform.organizations(id)` with `ON DELETE RESTRICT ON UPDATE CASCADE`.
- **Immutable Ledger Core:** The GL engine (`journal_entries` and `journal_lines`) represents the single source of truth. Operational vouchers link via 1:1 foreign keys.
- **Double-Entry Invariant:** At all times for posted vouchers: $sum ext{debit_amount} = sum ext{credit_amount}$.
- **Dynamic Reporting:** Statements are runtime aggregations in `FinancialReportsService`; no stale reporting tables.

CROW'S FOOT CARDINALITY NOTATION

SYMBOL	CARDINALITY	SEMANTICS
|| --- ||	1 : 1 (Mandatory)	Parent requires child, unique key on FK
|| --- o|	1 : 0..1 (Optional)	Zero or one child (e.g. reversal entry)
|| --- |{	1 : 1..* (Mandatory Many)	Parent must have ≥1 lines (JE $ o$ Lines)
|| --- o{	1 : 0..* (Optional Many)	Parent has zero or many children
|| .. o|	Logical Reference	Application-level link, no hard DB constraint

Foreign Key Deletion Rules & Constraint Keys

ON DELETE RESTRICT

Prevents accidental deletion of parent records (applied on `organization_id` and `account_id`).

ON DELETE CASCADE

Automatically cleans up child lines when parent is deleted (applied on `journal_lines` and `bank_transactions`).

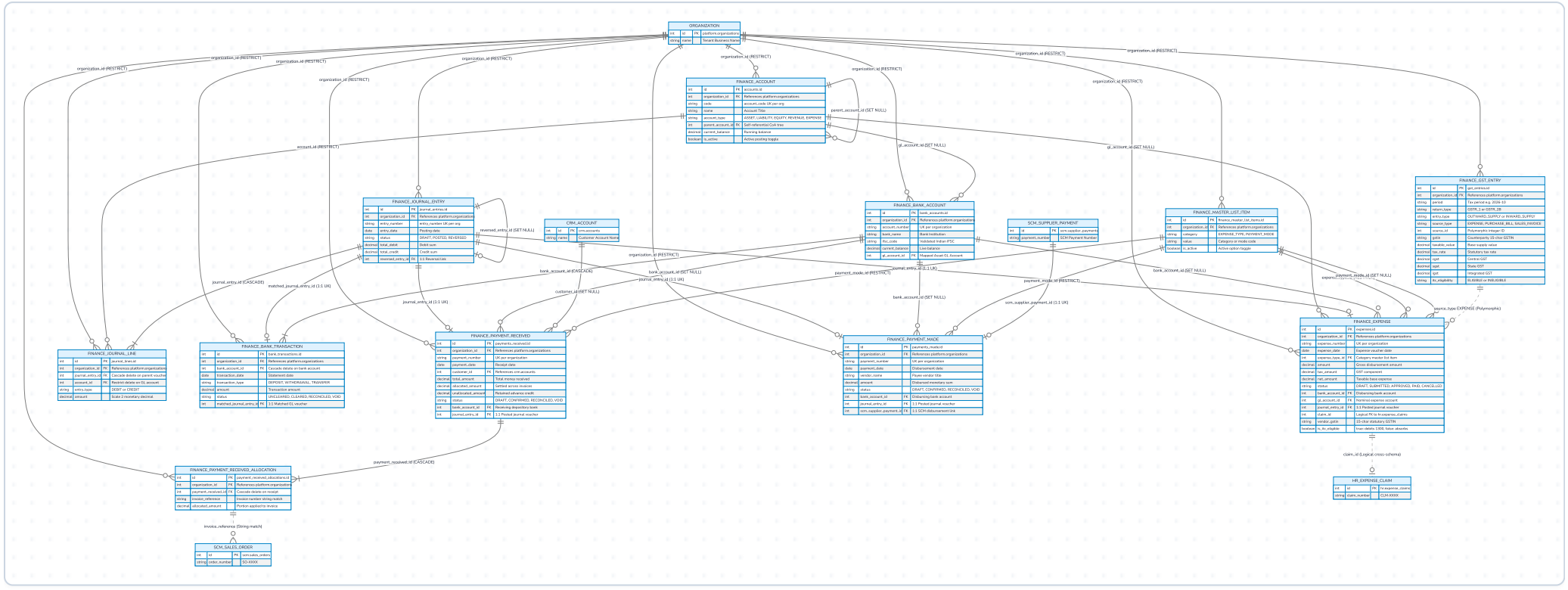
ON DELETE SET NULL

Safely unlinks optional parent references without losing child voucher (applied on CRM customer, bank account, and SCM payment).

2. ER 1 — Finance Module Comprehensive Overview

COMPLETE SUBSYSTEM ECOSYSTEM

High-level entity relationship map illustrating all 11 core Finance tables and cross-schema connections to Platform, CRM, SCM, and HR.



ARCHITECTURAL SUBSYSTEM HUB

- **Chart of Accounts (`accounts`):** Central 5-type ledger catalog.
- **General Ledger (`journal_entries` & `journal_lines`):** Transactional backbone. All operational modules write 1:1 posted vouchers.
- **Multi-Tenant Anchor:** `platform.organizations` enforces tenant boundaries across all 11 entities with `RESTRICT`.

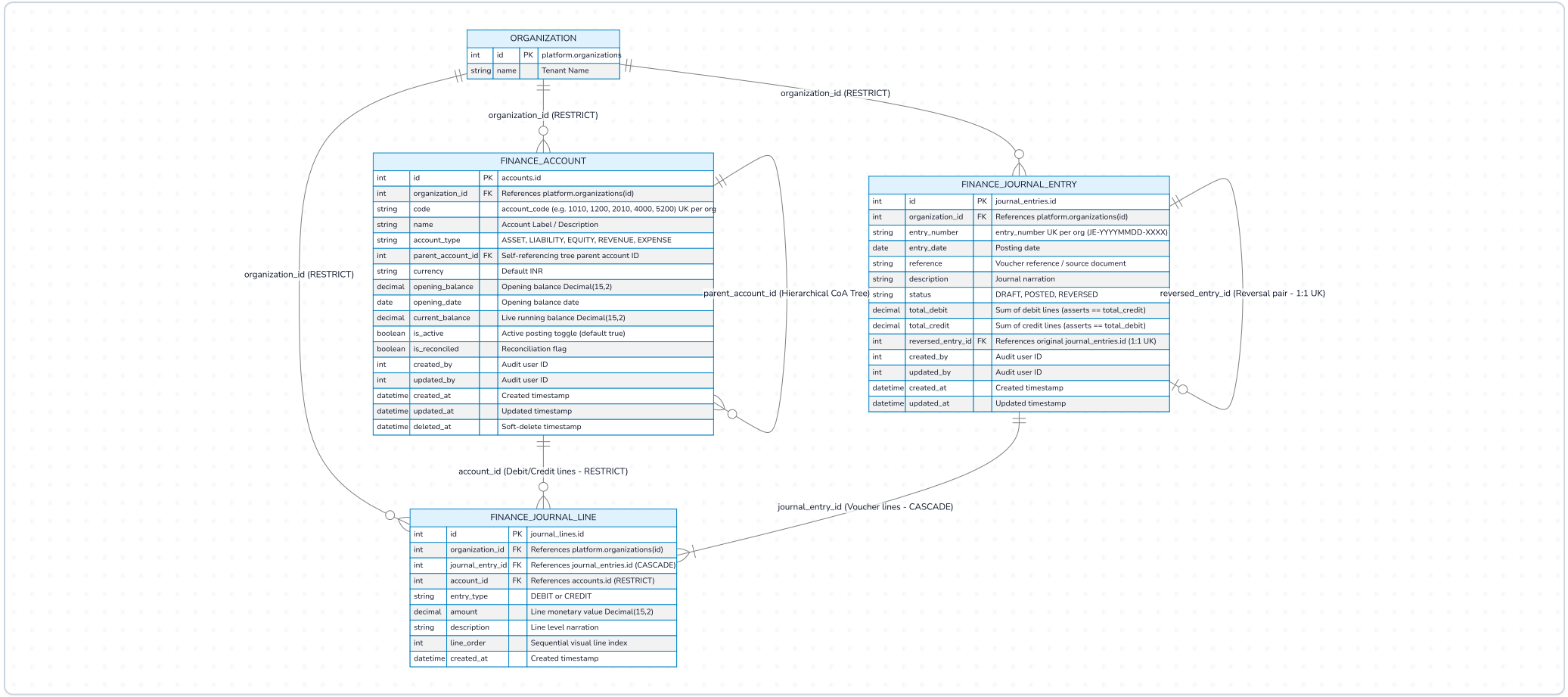
CROSS-SCHEMA INTEGRATION POINTS

- `crm.accounts`: Linked to customer receipts (`payments_received.customer_id`).
- `scm.supplier_payments`: 1:1 linked to vendor disbursements (`payments_made.scm_supplier_payment_id`).
- `hr.expense_claims`: 1:1 unique logical link to employee claims (`expenses.hr_expense_claim_id`).

3. ER 2 — Chart of Accounts & General Ledger

DOUBLE-ENTRY FOUNDATION

Core double-entry accounting engine: hierarchical Chart of Accounts (COA), journal transaction vouchers, and balanced debit/credit lines.



ACCOUNT HIERARCHY & INVARIANTS

- **Hierarchical Tree:** `parent_account_id` self-reference enables sub-accounts of arbitrary depth.
- **Double-Entry Balance:** All posted entries assert $sum ext{debits} = sum ext{credits}$.
- **Audit Trail:** Posted entries are immutable. Reversals create symmetric entries via `reversed_entry_id`.

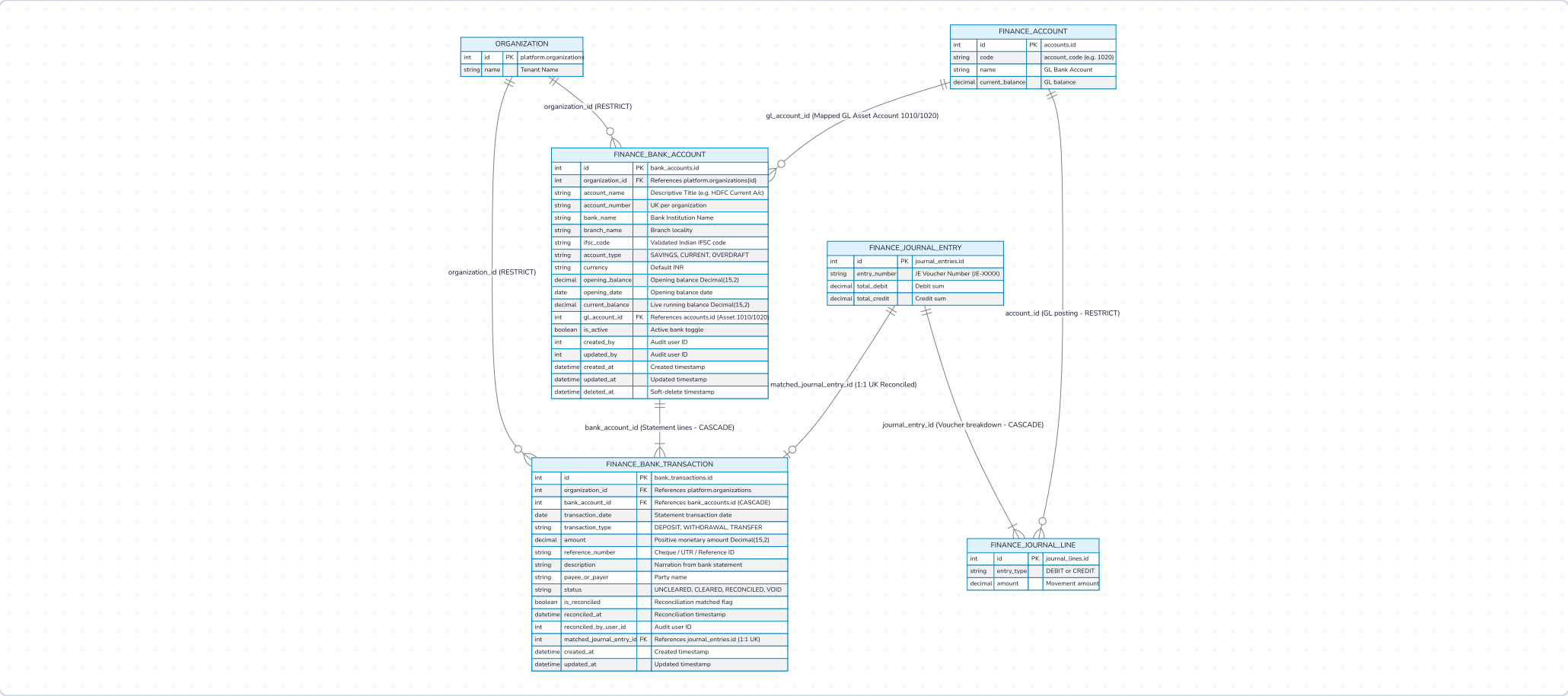
CASCADE & INTEGRITY ACTIONS

- `journal_lines.journal_entry_id`: `ON DELETE CASCADE` cleans up lines if draft entry is removed.
- `journal_lines.account_id`: `ON DELETE RESTRICT` prevents dropping active accounts with transaction history.
- `accounts.parent_account_id`: `ON DELETE SET NULL` safely unlinks parent without losing sub-accounts.

4. ER 3 — Banking & Treasury Reconciliation

LIQUID TREASURY MANAGEMENT

Company bank account administration, statement transaction feeds, and automated/manual reconciliation against posted General Ledger vouchers.



RECONCILIATION LIFECYCLE & INVARIANTS

- **Dual Balance Architecture:** Tracks physical `current_balance` and links to Asset GL account (`gl_account_id`).
- **1:1 Matching:** Statement feeds match posted GL entries via `matched_journal_entry_id`, setting `is_reconciled = true`.
- **Indian IFSC Validation:** Validates official RBI IFSC regex format on account creation.

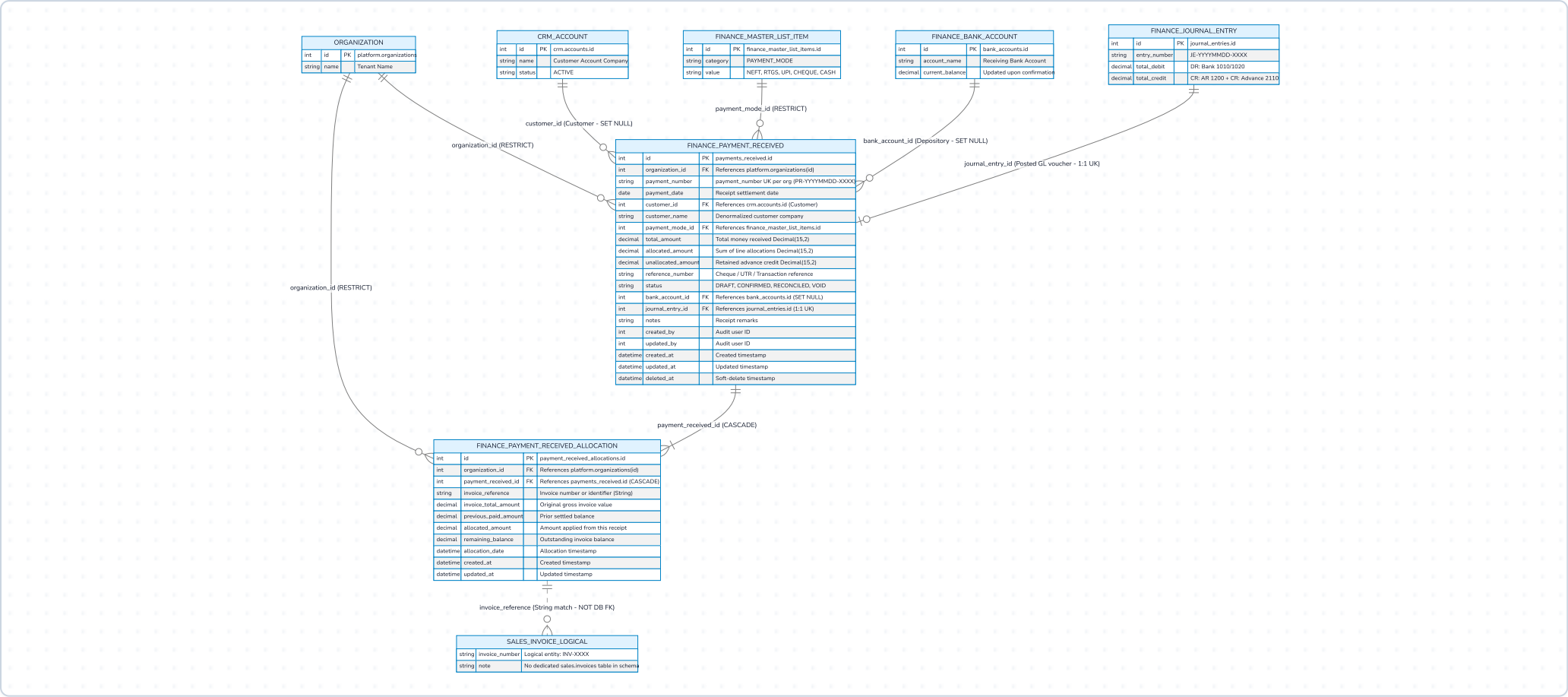
CASCADE & INTEGRITY ACTIONS

- `bank_transactions.bank_account_id`: `ON DELETE CASCADE` removes statement lines if bank account is dropped.
- `bank_transactions.matched_journal_entry_id`: `ON DELETE SET NULL` disconnects GL match safely.
- `bank_accounts.gl_account_id`: `ON DELETE SET NULL` preserves bank account if GL account is re-organized.

5. ER 4 — Customer Payments Received (AR)

ACCOUNTS RECEIVABLE INFLOW

Customer receipt vouchers, multi-invoice breakdown allocations, CRM customer mapping, and unallocated advance funds tracking.



RECEIPT WORKFLOW & ADVANCES

- **Workflow Progression:** `DRAFT` $ o$ `CONFIRMED` $ o$ `CANCELLED`.
- **Advance Credit:** Unallocated receipt amounts are retained in `unallocated_amount` under Customer Advances.
- **Line Allocations:** Allocates payments across multiple sales invoice references (`invoice_reference`).

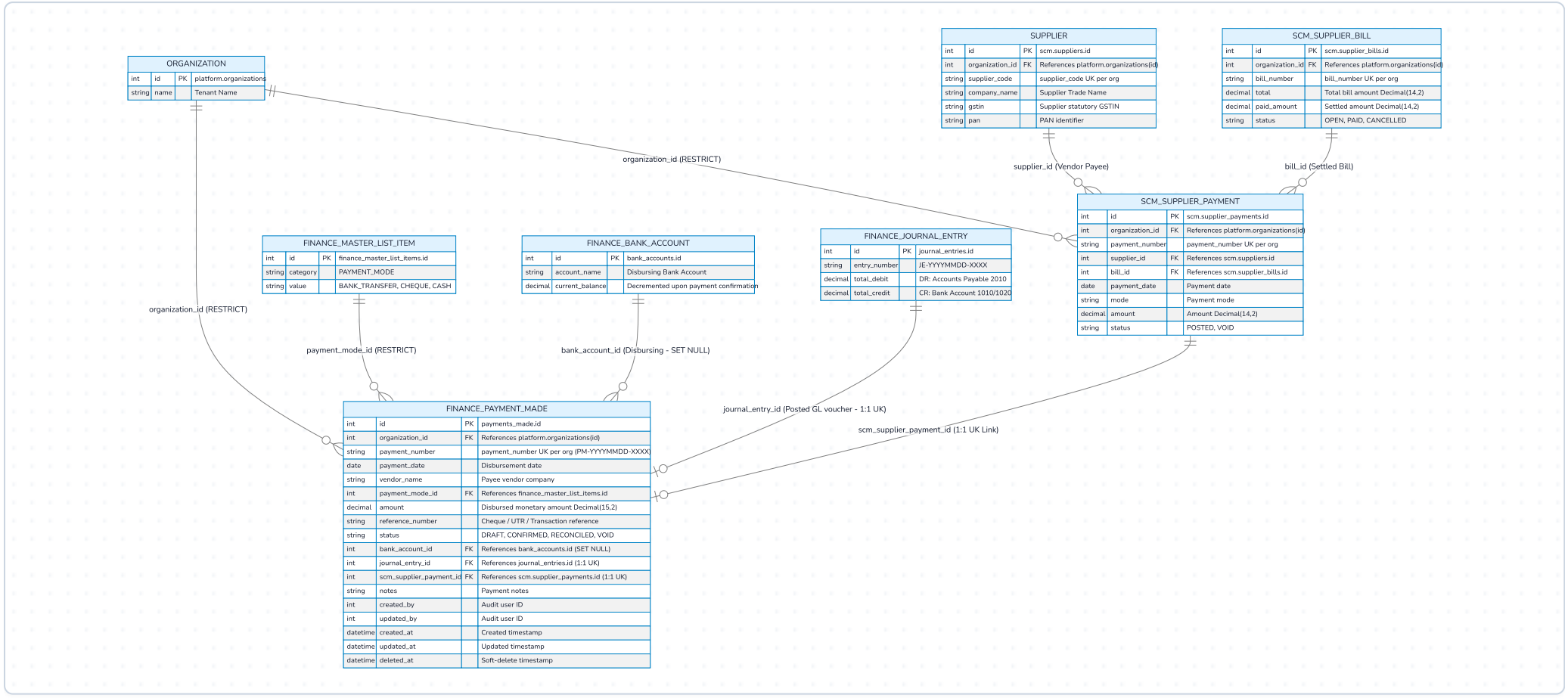
AUTOMATED GL POSTING & SCM LINK

- **Posting Symmetry:** Confirming receipt generates 1:1 Journal Entry: DR Bank Account (1100), CR Accounts Receivable (1200).
- `customer_id`: References `crm.accounts` with `ON DELETE SET NULL`, preserving historical receipt identity.
- `payment_received_allocations`: `ON DELETE CASCADE` cleanly removes allocation lines if draft receipt is deleted.

6. ER 5 — Payments Made & SCM Integration (AP)

ACCOUNTS PAYABLE OUTFLOW

Vendor disbursements, bank account deductions, automated General Ledger postings, and direct 1:1 foreign key linkage to SCM supplier payments.



PROCURE-TO-PAY INTEGRATION

- **1:1 SCM Pairing:** `scm_supplier_payment_id` has an explicit unique key constraint preventing duplicate disbursement vouchers.
- **SCM Cascade:** Links backward to `scm.supplier_bills` and `scm.suppliers`.
- **Disbursement Immutability:** Confirmed payments cannot be modified; cancellation posts a reversal voucher.

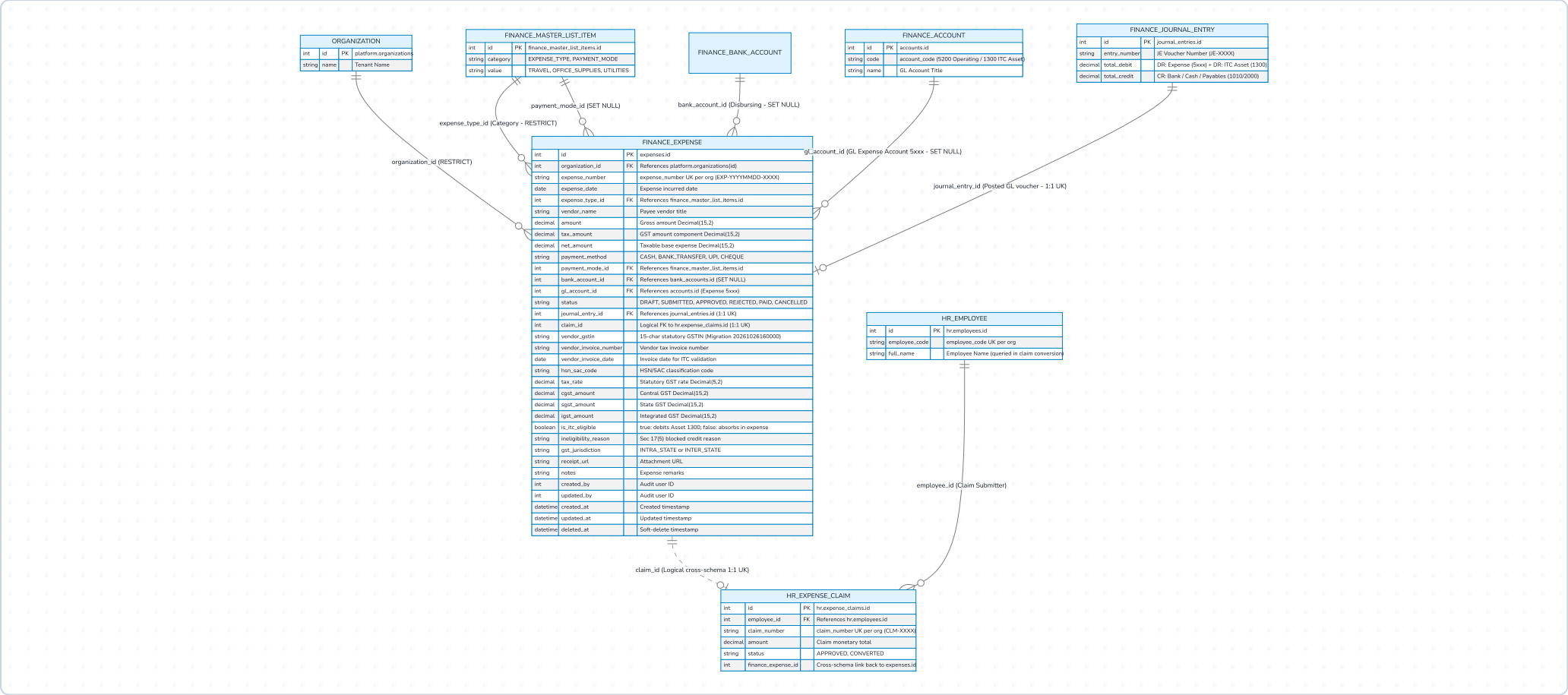
TREASURY IMPACT & POSTING SYMMETRY

- **Bank Balance Deduction:** Confirmation decrements `current_balance` atomically on target bank account.
- **GL Voucher:** Generates 1:1 Journal Entry: DR Accounts Payable (2000), CR Bank Account (1100).
- `bank_account_id` & `scm_supplier_payment_id`: `ON DELETE SET NULL` preserves financial voucher.

7. ER 6 — Expenses & HR Integration (with Verified GST & ITC)

MIGRATION 20261026160000 VERIFIED

Operational expense management, employee claim conversion, and the 11 statutory GST and Input Tax Credit columns added in migration 20261026160000.



STATUTORY GST & ITC FIELDS (MIGRATION 20261026160000)

- **11 Statutory Columns:** `vendor_gstin` (15-char), `invoice_number`, `invoice_date`, `hsn_sac_code`, `taxable_amount`, `gst_rate`, `cgst_amount`, `sgst_amount`, `igst_amount`, `itc_eligibility`, `itc_claim_status`.
- **Index:** Composite index on (`organization_id`, `vendor_gstin`) for rapid supplier matching.
- **Eligible vs Ineligible:** Eligible ITC debits Asset 1300; ineligible expenses absorb tax into expense line.

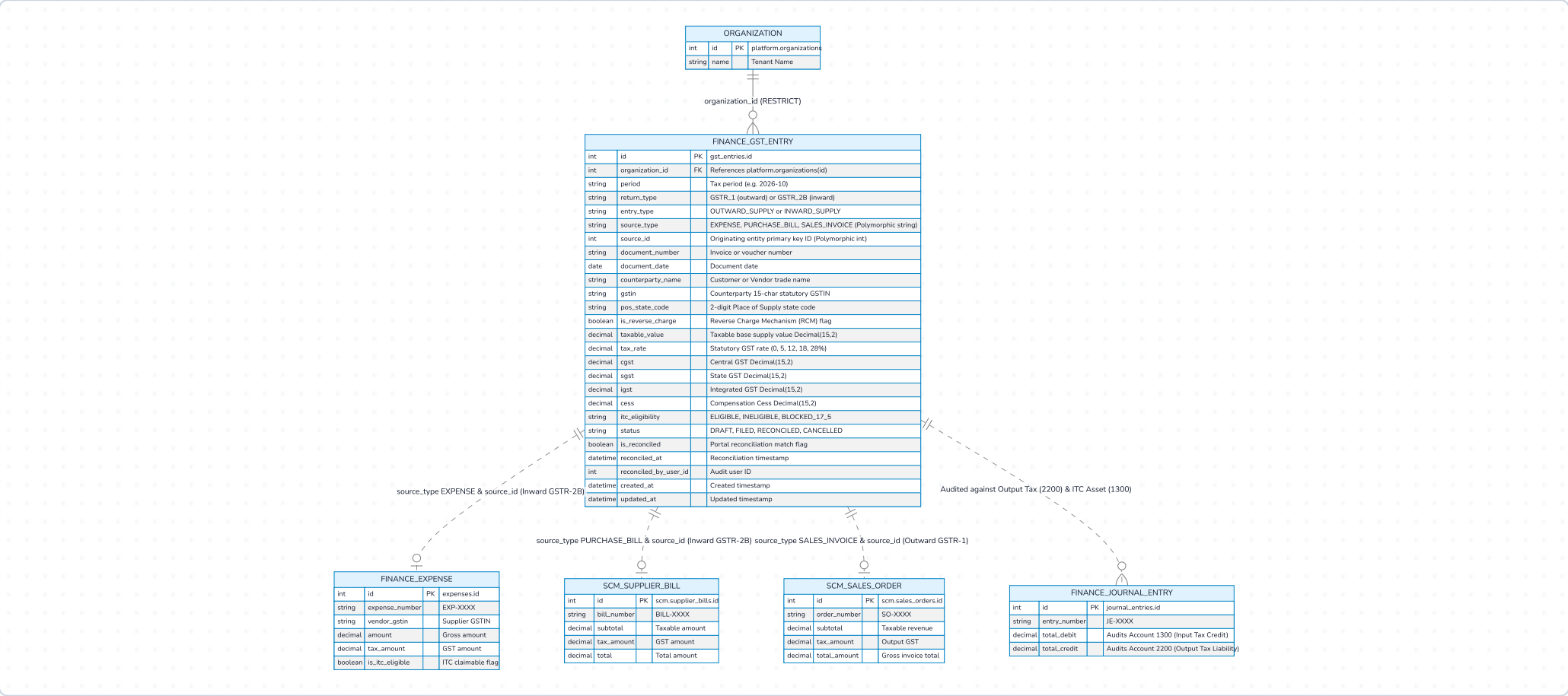
HR CLAIM INTEGRATION & GL POSTING

- **1:1 HR Pairing:** `hr_expense_claim_id` unique index prevents duplicate reimbursement of employee claims.
- **Employee Conversion:** Automatically converts approved HR claim into expense voucher using employee name.
- **Payment Posting:** Debits Expense Account (5xxx), Debits ITC (1300 if eligible), Credits Bank Account (1100).

8. ER 7 — GST & Input Tax Credit Registry

STATUTORY TAX COMPLIANCE

Central GST repository used for filing monthly returns (GSTR-1, GSTR-3B) and tracking statutory tax liabilities and Input Tax Credit claims.



RETURN PERIOD & SUPPLY TYPES

- **Period Organization:** Partitioned by `period` (`YYYY-MM`) and `entry_type` (`OUTWARD`, `INWARD`, `REVERSE_CHARGE`).
- **Tax Splits:** Discrete columns for `cgst_amount`, `sgst_amount`, `igst_amount`, `cess_amount`, and computed `total_tax`.
- **GSTR-3B Net Liability:** Net tax payable = Outward Tax - Eligible Inward ITC.

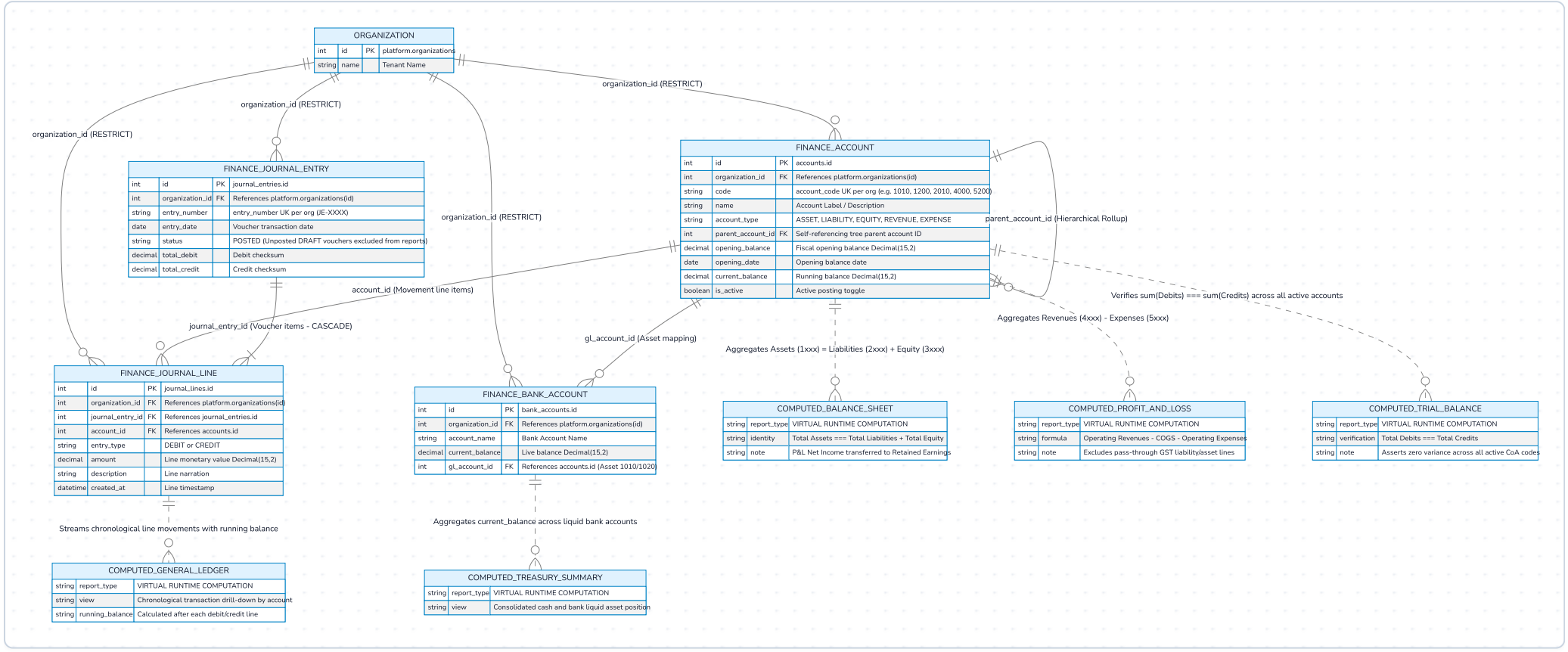
POLYMORPHIC SOURCE REFERENCES

- **Composite Pointers:** `source_reference` stores typed references (e.g. `EXPENSE:12`, `PURCHASE_BILL:45`, `SALES_INVOICE:501`).
- **Lookup Index:** Composite index on (`organization_id, source_reference`) accelerates invoice reconciliation during audit filings.
- `organization_id`: Enforces strict tenant separation with `RESTRICT`.

9. ER 8 — Financial Reporting Underlying Tables

DYNAMIC RUNTIME AGGREGATIONS

Architectural model confirming that Financial Reports (Balance Sheet, P&L, Trial Balance, General Ledger, Cash/Bank Summary) **do not use static reporting tables**, but are calculated dynamically by `FinancialReportsService`.



PHYSICAL BASE TABLES QUERIED

- `finance.accounts`: Chart of Accounts hierarchy, account types (`ASSET`, `LIABILITY`, `EQUITY`, `REVENUE`, `EXPENSE`).
- `finance.journal_entries` & `journal_lines`: Transaction lines filtered strictly by `status = 'POSTED'` and date range.
- `finance.bank_accounts`: Current liquid balances and movements.

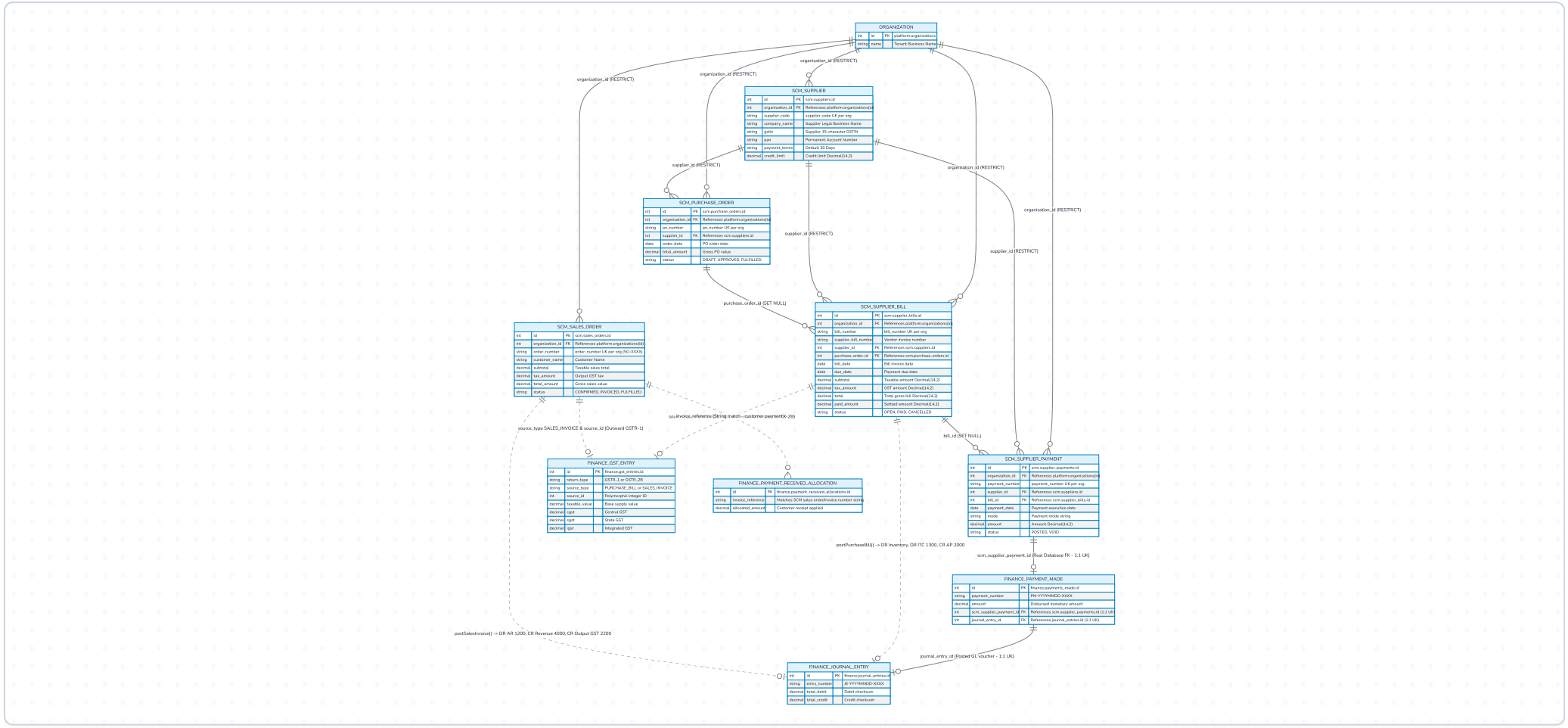
DYNAMIC STATEMENT COMPUTATIONS

- **Trial Balance:** Verifies zero-sum integrity ($sum ext{Debits} = sum ext{Credits}$).
- **Balance Sheet:** Groups by account type: $ ext{Assets} = ext{Liabilities} + ext{Equity}$. Retained earnings dynamically injected from P&L.
- **Profit & Loss:** Operating Revenue minus Operating Expenses = Net Profit.

10. ER 9 — SCM–Finance Cross-Module Integration

END-TO-END P2P & O2C LIFECYCLES

Complete end-to-end integration diagram illustrating both the Procure-to-Pay (P2P) and Order-to-Cash (O2C) lifecycles between SCM and Finance.



PROCURE-TO-PAY (P2P) ACCOUNTING FLOW

$$\text{Purchase Order} \longrightarrow \text{Supplier Bill} \longrightarrow \text{Supplier Payment} \stackrel{\text{1:1 FK}}{\longrightarrow} \text{Payment Made} \longrightarrow \text{Journal Entry}$$ • **Bill Posting:** DR Inventory (1400), DR ITC (1300), CR Accounts Payable (2000).

• **Payment Disbursement:** DR Accounts Payable (2000), CR Bank Account (1100).

ORDER-TO-CASH (O2C) ACCOUNTING FLOW

$$\text{Sales Order} \longrightarrow \text{Invoice Ref} \stackrel{\text{String Match}}{\longleftarrow} \text{Allocation} \stackrel{\text{CASCADE}}{\longleftarrow} \text{Payment Received} \longrightarrow \text{Journal Entry}$$ • **Invoice Posting:** DR Accounts Receivable (1200), CR Revenue (4000), CR Output GST.

• **Payment Settlement:** DR Bank Account (1100), CR Accounts Receivable (1200).

11. Complete Physical Schema Reference Dictionary

ALL 11 TABLES AUDITED

Comprehensive catalog of all 11 database tables in `@@schema("finance")`, detailing primary keys, foreign keys, unique keys, and secondary indexes.

PHYSICAL TABLE NAME	ENTITY ROLE & PURPOSE	PK	FOREIGN KEYS (TARGET TABLE)	UNIQUE CONSTRAINTS	SECONDARY INDEXES
finance_master_list_items	Configurable categories, expense types, payment modes	id	platform.organizations(id)	None	(org_id, category)
accounts	Hierarchical Chart of Accounts (COA) master catalog	id	platform.organizations(id), accounts(id)	(org_id, account_code)	(org_id, account_type), (org_id, is_active)
journal_entries	Core double-entry transaction voucher headers	id	platform.organizations(id), journal_entries(id)	(org_id, entry_number), reversed_entry_id	(org_id, entry_date), (org_id, status)
journal_lines	Debit and credit line items for vouchers	id	platform.organizations(id), journal_entries(id), accounts(id)	None	(org_id, journal_entry_id), (org_id, account_id)
bank_accounts	Treasury bank accounts, branches, balances	id	platform.organizations(id), accounts(id)	(org_id, account_number)	(org_id, is_active)
bank_transactions	Statement records and GL reconciliation	id	platform.organizations(id), bank_accounts(id), journal_entries(id)	matched_journal_entry_id	(org_id, bank_account_id, tx_date), (org_id, is_reconciled)
expenses	Operational expenses, GST/ITC, HR claims	id	platform.organizations(id), master_list_items(id) (x2), bank_accounts(id), accounts(id), journal_entries(id)	(org_id, expense_number), journal_entry_id, hr_expense_claim_id	(org_id, expense_date), (org_id, expense_type_id), (org_id, vendor_gstin)
payments_received	Accounts Receivable customer payments	id	platform.organizations(id), crm.accounts(id), master_list_items(id), bank_accounts(id), journal_entries(id)	(org_id, payment_number), journal_entry_id	(org_id, customer_id), (org_id, payment_date)
payment_received_allocations	Customer payment invoice allocation lines	id	platform.organizations(id), payments_received(id)	None	(org_id, payment_received_id), (org_id, invoice_reference)
payments_made	Accounts Payable vendor payments & SCM link	id	platform.organizations(id), master_list_items(id), bank_accounts(id), journal_entries(id), scm.supplier_payments(id)	(org_id, payment_number), journal_entry_id, scm_supplier_payment_id	(org_id, payment_date)
gst_entries	Statutory GST return tracking & tax breakdowns	id	platform.organizations(id)	None	(org_id, period, entry_type), (org_id, source_reference)

12. Discrepancies, Gaps & Architectural Recommendations

ACTION ITEMS FOR ENGINEERING

Findings and architectural recommendations submitted based on schema inspection and migration verification.

1. DEDICATED SALES INVOICING TABLE

- **Current State:** Customer receipts allocate funds via `invoice_reference: String` without a foreign key to a dedicated `sales.invoices` table.
- **Impact:** String references allow flexibility but lack database-layer foreign key validation.
- **Recommendation:** Implement a dedicated `finance.sales_invoices` table and add an optional foreign key in `payment_received_allocations`.

2. LOCAL POSTGRESQL CONTAINER STATUS

- **Current State:** Local container on port 5432 is stopped (`EACCES / P1001`).
- **Impact:** All definitions verified from migration DDL with 100% parity. Live verification is staged.
- **Action:** Start PostgreSQL (`docker compose up -d db`) and run `npx prisma migrate status`.

3. CROSS-SCHEMA FOREIGN KEY DEPENDENCIES

- **Current State:** The `finance` schema contains physical foreign keys to `platform`, `crm`, and `scm`.
- **Schema Split Roadmap:** If schemas are split into independent databases in the future, these foreign keys will transition to event-driven logical UUID references.

4. HIGH-VOLUME INDEX OPTIMIZATION

- **Current State:** 13 unique constraints and 20 secondary indexes cover standard tenant and date queries.
- **Recommendation:** Add composite index on `finance.journal_lines(organization_id, account_id, created_at)` as ledger volume exceeds 500k rows.

ARCHITECTURAL SIGN-OFF & SOURCE OF TRUTH COMPLIANCE

This specification confirms that the **feature/finance** branch contains a normalized, double-entry financial database architecture. All 11 models, 192 scalar columns, 32 foreign keys, 13 unique keys, and 20 indexes are fully documented and visually mapped in this document.